

LKA Invoices

Source code guide: what each main file does

Lingeshwar Kaparathi & Associates · Code: github.com/adi-repo-ai/ca-invoice-app (branches main and staging) · Updated October 2026

How the pieces fit. The website (React, in `src/`) runs in the browser. **Pages** show screens and call the **data layer** (`src/data/`), which reads and writes Firestore directly; every write is checked by `firestore.rules`. Pure calculations (GST, money, numbering) live in `src/lib/`. PDFs are drawn in the browser by `src/pdf/`. The few actions a browser must not be trusted with (roles, deleting accounts, first admin, numbering reset) go through `src/api.ts` to the **Netlify Functions** in `netlify/functions/`, which use the secret Firebase admin key. Money is always whole **paise** (₹1 = 100) to avoid rounding errors.

1. App shell

File	What it does
<code>src/main.tsx</code>	Starting point: mounts the app with the router and the sign-in provider.
<code>src/App.tsx</code>	All page addresses (routes) and the guards: signed-in only, admins only, revenue (owner) only. Shows the TEST SITE banner.
<code>src/firebase.ts</code>	Connects to Firebase (sign-in + Firestore) using the <code>VITE_FIREBASE_*</code> values; switches to the local emulators in development.
<code>src/auth.tsx</code>	Who is signed in, their role (ADMIN/STAFF) and permissions (revenue <code>fin</code> , payments <code>pay</code>) from the signed token; 30-minute idle sign-out; <code>useActor()</code> for audit entries.
<code>src/api.ts</code>	Calls a Netlify Function with the user's sign-in token attached.
<code>src/settings-context.tsx</code>	Loads firm settings once per visit and shares them with every page.
<code>src/theme.ts</code>	Light / dark / follow-device theme, remembered per device.

2. Pages (`src/pages/`)

File	Screen and main logic
<code>Login.tsx</code>	Google sign-in (main), email + password (fallback), forgot password, "No access yet" page with the one-time setup code.
<code>Home.tsx</code>	Dashboard: today panel, nudges, revenue tiles, 6-month chart and goal ring (owners), weekly summary, recent activity, milestones, personal tasks / events / notes / links, getting-started checklist.
<code>Invoices.tsx</code>	Invoice list: status tabs with counts, client and date filters, Newest / Oldest sort, hide amounts, CSV export (owners), delete one / delete all (admins).
<code>InvoiceEdit.tsx</code>	Create or edit a draft: client, service lines, reimbursements, live totals; save draft or issue (gets the next number); duplicate via <code>?from=</code> .
<code>InvoiceView.tsx</code>	One invoice: view / download PDF, email / WhatsApp, payment reminder, mark as paid and receipt (payments permission), cancel, duplicate, delete, history; back link to Invoices.
<code>Clients.tsx</code>	Client list: name search (also finds an invoice by number), tag filter, A-Z / newest, CSV import and template (admins), delete (admins).
<code>ClientEdit.tsx</code> / <code>ClientView.tsx</code>	Add or edit a client (tags, internal notes) / one client's details, statement of account PDF (payments permission) and invoice history.

Reports.tsx	Financial-year report (owners): billed vs received, collection rate, ageing, top clients, who owes.
Settings.tsx	Firm details, GST, bank / UPI, services (SAC), numbering (restart at 0001), goals and message templates, branding; Test site: sample data card (owners, test addresses only).
Users.tsx	Who has access: add a user with an access level, owner-only role / permission switches, disable, delete; read-only for non-owners.
Activity.tsx / Backup.tsx	Audit log of who did what (admins) / download a full JSON backup (owners).

3. Shared components (src/components/)

File	What it does
Layout.tsx	Header, menu (shows only what the person may use), search, bell, keyboard shortcuts, mobile "+" button, footer.
ui.tsx	Building blocks used everywhere: Button, Card, Field, Alert, PageHeader (with back link), Money, StatusBadge, Switch, Stat, Empty, Skeleton, Tip, OverdueBadge, and <code>errorMessage()</code> (friendly error text).
InvoiceTable.tsx	Invoice rows with status stripe, overdue badge and row actions (view, download, email, WhatsApp, remind, duplicate, delete).
ClientPicker.tsx / ClientFields.tsx	Search-as-you-type client chooser / client details typed directly on an invoice.
CommandPalette.tsx	Ctrl+K search: clients, invoice numbers, pages.
NotificationBell.tsx	Overdue invoices, tasks due, today's events, old drafts.
Dialog.tsx, Charts.tsx, EnvBanner.tsx	In-app confirm boxes; small bar / ring charts; the yellow "TEST SITE" / emulator banner.
SettingsTabs.tsx, ThemeToggle.tsx, icons.tsx	Settings sub-menu; day/night switch; icons.

4. Data layer (src/data/) — every database read and write

File	What it does
invoices.ts	The invoice engine: <code>createDraft</code> , <code>updateDraft</code> , <code>issueInvoice</code> (transaction: next number, today's date, firm snapshot), <code>recordPayment</code> , <code>cancelInvoice</code> , <code>deleteInvoices</code> ; lists, counts, totals, overdue, search by number. Each change writes an audit entry in the same step.
clients.ts	Add / edit / delete clients; name-prefix search, newest list, tag filter (with a fallback while its index builds), count.
settings.ts	Load and save the firm settings document.
audit.ts	Append-only activity log: add entries, list recent / per invoice, describe them in words.
home.ts	Each person's private tasks, events, notes and links.
sampleData.ts	Test site only: creates 6 TEST clients and 10 invoices through the normal functions above.

5. Logic library (src/lib/) — pure functions, unit-tested

File	What it does
------	--------------

tax.ts	GST engine: CGST + SGST (same state) or IGST (other state) or no GST; line totals, rounding, grand total, amount in words.
money.ts / words.ts	Rupees ↔ paise, Indian number formatting (1,00,000) / amount in words (thousand, lakh, crore).
fy.ts	Indian financial year (April–March), IST date, invoice number format (e.g. LKA/2026-27/0001), date maths.
types.ts / defaults.ts	Shapes of every record / starting settings until the owner saves their own.
validation.ts / states.ts	GSTIN, PAN, IFSC, email, phone checks / GST state codes.
messages.ts / send.ts	Message text from templates, UPI pay link / share or download PDF and open WhatsApp or email.
report.ts, csv.ts, clientImport.ts	Report numbers; safe CSV export; CSV client import.
siteEnv.ts	<code>isTestHost()</code> : true only on Netlify test addresses (with "--"); drives the banner and sample-data card.
celebrate.ts, image.ts, defaultLogo.ts	Confetti; logo resizing; default logo.

6. PDFs (src/pdf/) — drawn in the browser

File	What it does
InvoicePdf.tsx	A4 invoice: header, bill-to, service and reimbursement tables, totals, amount in words, payment details with UPI QR (or "Received with thanks" when paid), signature, terms, PAID / DRAFT / CANCELLED stamp.
ReceiptPdf.tsx / StatementPdf.tsx	Payment receipt / client statement of account with running balance.
common.tsx / generate.ts	Shared header, footer and styles / turns a layout into a downloadable file with the QR code.

7. Server functions (netlify/functions/) — hold the secret admin key

File	What it does
activate.ts	On first Google sign-in: if an owner added this email, gives the role and access level; otherwise refuses.
admin-users.ts	Owners only: create, change role, set permissions, disable / enable, delete users, set password.
claim-admin.ts	One-time: the first person becomes owner with the <code>SETUP_CODE</code> .
reset-numbering.ts	Restart this year's numbering at 0001 once no numbered invoice of the year is left.
_shared/admin.ts, _shared/http.ts	Admin SDK connection from env vars; token check, role / owner check, default permissions, error replies.

8. Security, configuration, scripts and tests

File	What it does
firestore.rules	The real security. Who may read / write each collection; invoices numbered without gaps; issued invoices locked; payments need the payments permission; deletes admin-only; invites owner-only; audit log append-only.

<code>firebase.indexes.json</code>	Search / sort indexes (status + date, client + date, tags + name, ...). Deployed from Cloud Shell.
<code>netlify.toml</code> , <code>firebase.json</code>	Build settings, security headers, page routing / which rules and indexes files Firebase deploys.
<code>.env.example</code> , <code>package.json</code>	List of settings needed (no real values) / libraries and commands (<code>npm run build</code> , <code>test:all</code> , <code>seed:test</code> ...).
<code>scripts/</code>	<code>create-admin</code> (make an owner; <code>--key</code> file option), <code>seed-emulator</code> (local demo), <code>seed-test-data</code> (test project only; refuses live).
<code>tests/emulator/</code> , <code>src/**/*.test.ts</code>	93 automated checks: security rules, server functions, numbering, GST, money, words, CSV, reports, test-site detection.

9. Where to change what

To change...	Edit
Invoice PDF look	<code>src/pdf/InvoicePdf.tsx</code> (shared header/footer: <code>src/pdf/common.tsx</code>)
GST / totals	<code>src/lib/tax.ts</code> (+ its tests)
Who can do what	<code>firebase.rules</code> + screens (<code>src/App.tsx</code> , pages) + <code>netlify/functions/</code> ; then deploy rules
A screen's layout / buttons	the file in <code>src/pages/</code>
Firm details, services, templates	No code: Settings in the portal